BIOLOGY AND TECHNOLOGY

PYQ • 2021 - 2026

1. 2026 **1 MARK**

In gene therapy using stem cells, viruses are mainly used as vectors because :

(a) Help in inserting the active gene into stem cells.

(b) Destroy the defective gene in defective cells.

(c) Increase the number of stem cells in the body.

(d) Convert body cells into stem cells.

2. 2026 **2 MARKS**

Mention the advantages of the following technological possibilities in environmental conservation.

(a) Bioremediation

(b) Cryopreservation

3. 2026 **2 MARKS**

(A) The steps involved in recombinant DNA technology are illustrated below.

(a) Identify the DNA into which the cut human gene is ligated.

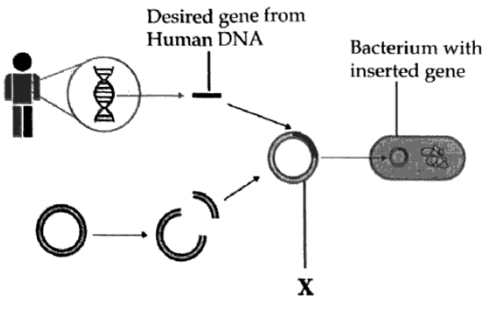
(b) What happens when 'X' is inserted into the host cell ?

OR

(B) CRISPR is a modern technology used for precise DNA editing.

(a) Explain the role of the following in this technology : Guide RNA ; Cas9 enzyme

(b) How does CRISPR gene editing help to change the characteristics of an organism ?



4. 2025 **1 MARK**

Some diseases and proteins produced through Genetic engineering for their treatment are given. Choose the correct pair.

(a) Insulin – Growth disorders (b) Interferons – Viral diseases

(c) Endorphins – Diabetes (d) Somatotropin – Pain

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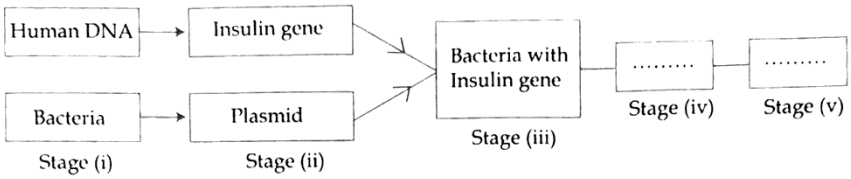
5. 2025 **3 MARKS**

Analyse the illustration indicating the stages of Insulin production through Genetic Engineering and answer the following questions.

(a) Name the enzymes used in this process.

(b) What is the importance of plasmid in this process ?

(c) Complete the stages indicated by (iv) and (v).



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6. 2024 **1 MARK**

Correct the mistakes if any in the underlined part of the given statements.

(a) The complete genetic material present in an organism is called its genome.

(b) Gene mapping helps to identify the location of a particular gene in the DNA.

(c) The technology of testing the arrangement of nucleotides is gene therapy.

(d) Interferons are the proteins used for the treatment of viral diseases.

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7. 2024 **3 MARKS**

Analyse the steps in the production of human insulin through genetic engineering and answer the questions.

Step 1. Cutting of insulin gene from human DNA

Step 2. Joining insulin gene with plasmid

Step 3. Plasmid with ligated insulin gene is inserted into bacterial cell

Step 4. Bacteria multiply in the culture medium

(a) Which are the enzymes used in step 1 and 2.

(b) What is the role of plasmid in this process ?

(c) Whether the succeeding generations of this bacterium have the ability to produce insulin. Why ?

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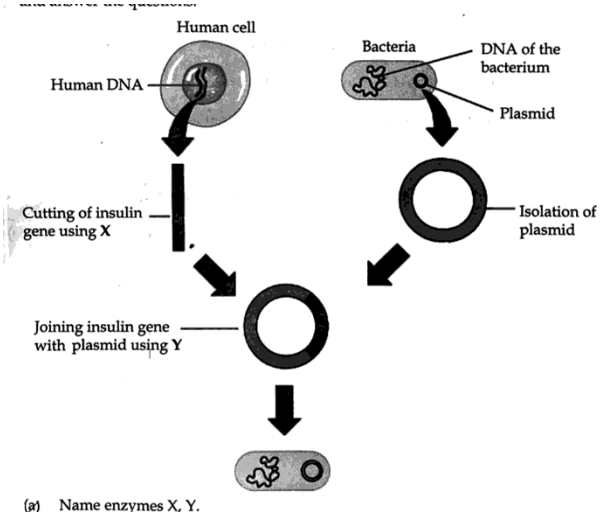
8. 2023 **3 MARKS**

Observe the illustration showing the production of insulin through Genetic engineering and answer the questions.

(a) Name enzymes X, Y.

(b) What is the role of plasmid in this process ?

(c) Does the succeeding generations of bacteria produced in this way have the ability to produce insulin ? Why ?



(a) Name enzymes X, Y.

9. 2022 **1 MARK**

The vector used in genetic engineering is :

(a) Ligase (b) Plasmid (c) tRNA (d) rRNA

10. 2022 **2 MARKS**

Analyse given statement and answer the following questions.

'Genetic engineering has made a great leap in the treatment of genetic diseases'.

(a) Name the treatment method indicated.

(b) What is the specific peculiarity of this treatment method ?

11. 2021 **1 MARK**

The protein produced through genetic engineering which is used against viral diseases :

(a) Insulin (b) Interferons (c) Endorphin (d) Somatotropin

12. 2021 **1 MARK**

Identify the word pair relation and fill in the blank.

Genetic scissors : restriction endonuclease

Genetic glue : _____

13. 2021 **2 MARKS**

Write any two uses of DNA fingerprinting technology.